

Dr. Shafqat Ali Siyyal

Date of birth: 03/04/1996

Nationality: Pakistani

Gender: Male

CONTACT

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ABOUT ME

Ph.D. in Control and Autonomous Systems, with research focused on modeling, control & optimization, and resilience of cyber-physical and autonomous systems. Experience includes linear and nonlinear system modeling, PID control, Model Predictive Control (MPC), and Robust MPC approaches. Research also involved observer-based detection of system anomalies caused by stochastic disturbances, faults, or intentional cyberattacks. Additionally, developed and investigated Software Rejuvenation (SWR) frameworks for improving system resilience, recovery, and safe operation under adverse conditions.

WORK EXPERIENCE

● **UNIVPM - Università Politecnica delle Marche** Ancona, Italy

Doctoral Researcher

01/11/2022 – 01/05/2026

- Conducting research on the security and resilience of cyber physical systems (CPS), with emphasis on autonomous control architectures
- Performing mathematical modeling and simulation of dynamic systems (e.g., quadrotor, four-tank system) using MATLAB/Simulink for system analysis and controller development
- Applying formal verification and analysis techniques including stability, invariant-set, and reachability analysis to evaluate system safety and performance bounds
- Designing and tuning controllers such as PID, LQR, pole-placement, MPC and feedback linearization, alongside observer schemes (Luenberger and Kalman) integrated with residual-based anomaly detection mechanisms
- working on a software rejuvenation framework to enhance runtime reliability and system safety while minimizing mission performance degradation

● **Universidad de Sevilla** Sevilla, Spain

Visiting Doctoral Researcher

01/05/2024 – 20/12/2024

- Collaborated on the development of robust MPC frameworks, including min-max MPC, tube-based MPC and explicit MPC, for enhancing control performance under uncertainties and adversarial conditions
- Formulated and solved convex and non-convex optimization problems for deriving optimal attack strategies from an adversarial perspective to evaluate system vulnerabilities
- Utilized MATLAB and optimization frameworks such as YALMIP, Gurobi, QP and fmincon solvers for controller and performance evaluation

● **Mehran University of Engineering & Technology, (MUET)** Jamshoro, Pakistan

Graduate Researcher

01/2022 – 07/2022

Outline: Towards Real Traffic based, Updated Dataset for Anomaly-based IDS

- Conducted a gap analysis of existing cybersecurity datasets to define requirements for a new host-based IDS dataset
- Proposed a virtual network testbed using Linux/Unix machines to launch and capture real-world attack traffic (e.g., DoS, DDoS, U2L, R2L)
- Trained and evaluated multiple machine learning models on the newly captured dataset to benchmark anomaly detection performance

● **Quaid-e-Awam University of Engineering, Science & Technology, (QUEST)**

Nawabshah Nawabshah, Pakistan

Undergraduate Researcher

03/2018 – 09/2018

Outline: Analysis of Different Routing Protocols of Mobile Ad-hoc Network (MANET) over Different Scenarios

- Designed and implemented a realistic MANET simulation environment in NS-2 to analyze AODV, DSDV, and DSR routing protocols
- Evaluated protocol performance across diverse scenarios by varying network scale and node mobility speeds
- Analyzed and compared protocol efficiency based on key metrics: packet delivery ratio (PDR), throughput, packet drops, end-to-end delay, and normalized routing load

Pakistan International Airlines

Intern

19/06/2018 – 04/07/2018

- Gained foundational knowledge of aircraft communication systems, including radio wave principles and antenna functions
- Observed Air Traffic Control (ATC) operations, learning the procedures for maintaining aircraft surveillance and radar contact
- Became familiar with the processes used by ATCs to manage and separate aircraft activities

Pakistan Telecommunication Company LTD

Intern

26/06/2016 – 07/08/2016

- Assisted the network team in monitoring the NMS for fault reports and critical alarms
- Supported staff with basic configuration tasks on switching equipment under supervision

EDUCATION AND TRAINING

01/11/2022 – 01/05/2026 Bari, Italy

Doctor of Philosophy (PhD) in Autonomous Systems Politecnico di Bari & UNIVPM (Università Politecnica delle Marche, Ancona)

Field of study Department of Information Engineering | **Thesis** Model-based design for increasing reliability and safety of autonomous systems

01/2019 – 10/2022 Jamshoro, Pakistan

Master of Engineering in Communication Systems and Networks Mehran University of Engineering & Technology

Thesis: Analyzing ML based IDS over Real Traffic

Supervisor: Assoc Prof. Dr. Faheem Yar Khuhawar

Website <https://www.muett.edu.pk> | **Final grade** 3.74

01/2015 – 12/2018 Nawabshah, Pakistan

Bachelor of Engineering in Telecommunication Engineering Quaid-e-Awam University of Engineering Science & Technology

Thesis: Comparative study on the Performance of MANET Routing Protocols in Different Scenarios

Supervisor: Assoc Prof. Dr. Ahmed Arain

Website <https://quest.edu.pk> | **Final grade** 77%

LANGUAGE SKILLS

MOTHER TONGUE(S): Urdu

Other language(s):

English

Listening C1

Reading C1

Writing C1

Spoken production B2

Spoken interaction B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Control and optimization | robust MPC | formal analysis of control systems | MATLAB/Simulink | YALMIP | LaTeX

PUBLICATIONS

Journals & Conferences

1. **Siyyal, S.A.**, Freddi, A., Maestre, J.M., Romagnoli, R., Baldini, A. and Longhi, S., 2026. Condition-Based Software Rejuvenation Framework for Cyber-Physical Systems using a Dual-Monitor Architecture. In *23rd IFAC World Congress* (Busan, Republic of Korea) (Accepted).
2. **Siyyal, Shafqat A.**, et al. "A Constraints-aware Antagonistic Controller with Disturbance-adaptive Attacks." *IFAC-PapersOnLine* 59.11 (2025): 13-18. Retrieved from <https://www.sciencedirect.com/science/article/pii/S2405896325012807>
3. Cavanini, L., Felicetti, R., Ferracuti, F., Freddi, A., Longhi, S. and **Siyyal, S.A.**, 2024, September. Antagonistic Model Predictive Control for Quadrotor Cyber Attacks. In *2024 20th IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications (MESA)* (pp. 1-6). IEEE Retrieved from <https://ieeexplore.ieee.org/abstract/document/10704840>
4. **Siyyal, Shafqat Ali**, et al. "Development and validation of dataset for intrusion detection system over real traffic." *2022 25th International Symposium on Wireless Personal Multimedia Communications (WPMC)*. IEEE, 2022. Retrieved from <https://ieeexplore.ieee.org/abstract/document/10014923>
5. **Siyyal, S. A.**, Faheem Yar Khuawar, Erum Saba, Abdul Latif Memon, & Muhammad Raza Shaikh. (2022). Analyzing ML-Based IDS over Real-Traffic. *International Journal of Innovations in Science & Technology*, 4(3), 621–640. Retrieved from <https://journal.50sea.com/index.php/IJIST/article/view/305>
6. **S. Ali**, A. Ahmed and M. Raza, "Towards Better Routing Protocols for IoT," 2019 2nd International Conference on Computing, Mathematics and Engineering Technologies (iCoMET), 2019, pp. 1-5, doi: 10.1109/ICOMET.2019.8673456. Retrieved from <https://ieeexplore.ieee.org/document/8673456>

ADDITIONAL SKILLS

Transferable Skills

Research and Analysis, Communicating Results (Presentations/Academic writing), Persistence, Teamwork & Leadership skills, Independent or Self-starter

Technical Skills

- **Python** (Experience in Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn)
- **SQL** (Exposure to Data Analysis)
- **MATLAB** (Experience in Signal Processing & visualizations, Image Processing, Plotting of functions)
- **Cisco Packet Tracer, NS-2 & Wireshark** (Network simulations & its analysis, Traffic analysis)
- **Linux/Unix Environments** (Kali, Ubuntu, Metasploit-2)
- **C++** (requisite knowledge)